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Biochemistry: Dr. Nowrouzi

Biochemistry		
As the pKa of acid decreases, the acid will be:		۱
Weaker	الف	
Stronger	ب	
Converted to a neutral solution	ج	
None of them	د	
منبع:		

Biochemistry		
Which type of interaction mostly form between charged groups of biomolecules?		۲
Hydrophobic Interactions	الف	
Salt bridges	ب	
van der Waals Forces	ج	
Hydrogen bonds	د	
منبع:		

Biochemistry		
Which of the following acid/base pairs <u>cannot</u> act as natural buffers in living systems?		۳
H_2CO_3/HCO_3^-	الف	
$H_2PO_4^-/HPO_4^{2-}$	ب	
Histidine ⁺ /histidine	ج	
Glutamate ⁻ /Glutamic acid	د	
منبع:		

Biochemistry		
Which monosaccharide <u>cannot</u> form D and L isomerism?		۴
Dihydroxyacetone	الف	
Galactose	ب	
Fructose	ج	
Glyceraldehyde	د	
منبع:		

Biochemistry

Which carbohydrate can be excreted in the urine in poorly controlled diabetes mellitus?

۵

Mannose الف

Galactose ب

Ribose ج

Glucose د

منبع:

Biochemistry

Which structure is a derivative of monosaccharide and can be found in the structure of DNA

۶

Fucose الف

Mannose ب

Deoxyribose ج

Deoxyglucose د

منبع:

Biochemistry

Which derivative of monosaccharide is implicated in cataract formation?

۷

Phosphate esters of glucose الف

gluconic acid ب

Sorbitol ج

glucaric acid د

منبع:

Biochemistry

Which carbohydrate is used to determine the glomerular filtration rate?

۸

Inulin الف

Dextrin ب

Chondroitin sulfate ج

Uronic acids د

منبع:

Biochemistry

Which structure is a heteropolysaccharide and acts as a lubricant?

۹

Hyaluronic acid الف

Dextrin ب

Glycogen ج

Heparin د

منبع:



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Biochemistry		
Phospholipid acting as surfactant is.....		10
Cephalin	الف	
Phosphatidyl inositol	ب	
Lecithin	ج	
Phosphatidyl serine	د	
منبع:		

Biochemistry		
Dietary fats after absorption appear in the circulation as		11
HDL	الف	
VLDL	ب	
LDL	ج	
Chylomicron	د	
منبع:		

Biochemistry		
All the following have 18 carbon atoms <u>except</u> :		12
Linoleic acid	الف	
Linolenic acid	ب	
Arachidonic acid	ج	
Stearic acid	د	
منبع:		

Biochemistry		
HDL is synthesized in.....		13
Adipose tissue	الف	
Liver and intestine	ب	
Intestine	ج	
Liver	د	
منبع:		

Biochemistry		
Sphingosine is synthesized from		14
Palmitoyl CoA and Choline	الف	
linoleiyl CoA and ethanolamine	ب	
Acetyl CoA and choline	ج	
Palmitoyl CoA and serine	د	
منبع:		

3/5/14
14
0.6

Biochemistry			
Which of the following proteins is defective in Ehlers -Danlos syndrome?			20
Myoglobin	الف		
Elastase	ب		
Collagen	ج	☒	
Hemoglobin	د		
			منبع:

Biochemistry			
Which of the following correctly describes the Bohr effect?			21
Effect of temprature on the binding and release of oxygen by hemoglobin	الف		
Effect of 2,3-BPG on the binding and release of oxygen by hemoglobin	ب		
Effect of Carbon monoxide on the binding and release of oxygen by hemoglobin	ج		
Effect of pH and CO ₂ concentration on the binding and release of oxygen by hemoglobin	د	☒	
			منبع:

Biochemistry			
Which of the following reactions is catalyzed by hydrolase?			22
Cleavage of bonds by addition of water	الف	☒	
Intramolecular rearrangement of bonds	ب		
Transfer of group from one molecule to another	ج		
Cleavage of C-C and C-S bonds	د		
			منبع:

Biochemistry			
In non-competitive inhibition, what happens to V _{max} and K _m ?			23
K _m increased but V _{max} is constant	الف		
K _m increased but V _{max} decreased	ب		
V _{max} decreased but K _m is constant	ج	☒	
V _{max} decreased but K _m increased	د		
			منبع:

Biochemistry			
The general mechanism is that an enzyme acts by?			24
Increasing the free energies of the reactants	الف		
Reducing the activation energy	ب	☒	
Decreasing the free energies of the products	ج		
Increasing activation energy	د		
			منبع:

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Biochemistry		
Inhibition of dihydrofolate reductase by methotrexate falls into which category of inhibition?		25
Competitive inhibition	الف	
Irreversible inhibition	ب	
Un-competitive inhibition	ج	
Non-competitive inhibition	د	
		منبع:

Biochemistry		
Which of the following statement is not true about the K_m (Michaelis constant)?		26
Is a measure of the affinity of the enzyme for its substrate	الف	
Has a higher numerical value when the enzyme has high affinity for its substrate	ب	
Is the substrate concentration when the reaction rate is half of the maximum rate	ج	
The numerical value does not vary with enzyme concentration	د	
		منبع:

Biochemistry		
Which statement is incorrect about allosteric enzymes?		27
Display a sigmoidal dependence on substrate concentration	الف	
Frequently catalyze the committed step early in a pathway	ب	
Are almost always composed of multiple subunits	ج	
An allosteric substrate functions as a negative effector	د	
		منبع:

Biochemistry		
Thiamine deficiency causes all except :		28
Cheilosis	الف	
Ataxia	ب	
Nystagmus	ج	
Opthalmoplegia	د	
		منبع:

Biochemistry		
Vitamin which is excreted in urine is:		29
Vitamin C	الف	
Vitamin A	ب	
Vitamin D	ج	
Vitamin K	د	
		منبع:



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Biochemistry

In RNA molecule 'Caps'

- Allow tRNA to be processed
Are unique to eukaryotic mRNA
Occurs the 3' end of tRNA
Allow correct translation of prokaryotic mRNA

الف
ب
ج
د

35

منبع:

Biochemistry

In a DNA molecule, the guanosine content is 40%, and the adenine content will be

- 10% الف
20% ب
30% ج
40% د

36

منبع:

Biochemistry

Translocation of the newly formed peptidyl tRNA at the A site into the empty P site involves....

- EF-II, GTP الف
EF-I, GTP ب
EF-I, GDP ج
Peptidyl transferase, GTP د

37

منبع:

Biochemistry

The binding of prokaryotic RNA polymerase to promoter sites of genes is inhibited by the antibiotic:

- PuroThermycin الف
Streptomycin ب
Rifamycin ج
Terramycin د

38

منبع:

Biochemistry

Repressor binds to DNA sequence and regulates the transcription. This sequence is called

- Terminator الف
Operator ب
Attenuator ج
Anti terminator د

39

منبع:



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Biochemistry			
RNA primer is formed by the enzyme:			40
Ribonuclease	الف		
DNA polymerase I	ب		
Primase	ج		
DNA polymerase III	د		
منبع:			

Biochemistry			
Reverse transcriptase activity is present in the eukaryotic:			41
DNA polymerase α	الف		
DNA polymerase γ	ب		
Telomerase	ج		
DNA polymerase II	د		
منبع:			

Biochemistry			
The initiation site for transcription is recognized by.....			42
α -Subunit of DNA-dependent RNA polymerase	الف		
β -Subunit of DNA-dependent RNA polymerase	ب		
Rho factor	ج		
Sigma factor	د		
منبع:			

Biochemistry			
Novobicin inhibits the synthesis of			43
DNA	الف		
tRNA	ب		
mRNA	ج		
rRNA	د		
منبع:			

Biochemistry			
The first aminoacyl tRNA which initiates translation in prokaryotes is			44
Methionyl tRNA	الف		
Formylmethionyl tRNA	ب		
Tyrosinyl tRNA	ج		
Alanyl tRNA	د		
منبع:			



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Biochemistry		
Transcription of A, Y, and Z genes of the lac operon is prevented by.....		45
Lactose	الف	
Repressor	ب	☒
Allo-lactose	ج	
cAMP	د	
منبع:		

Biochemistry		
Very low-density lipoproteins are also known as.....		46
β -lipoproteins	الف	
Pre β -lipoproteins	ب	☒
α -lipoprotein	ج	
Broad β -lipoproteins	د	
منبع:		

Biochemistry		
Which of the following will form a palindromic sequence?		47
AGTCCTGA	الف	
GTTCCAAG	ب	
ATTGCAAT	ج	☒
GTTGGAAC	د	
منبع:		

Biochemistry		
The enzyme responsible for the removal of supercoiling in replicating DNA ahead of the replication fork is _____		48
Topoisomerase	الف	☒
Primase	ب	
DNA polymerase	ج	
Helicase	د	
منبع:		

Biochemistry		
Which of the following acts as a Lung surfactant?		49
Dipalmitoyl lecithin	الف	☒
Phosphatidyl serin	ب	
Diphosphatidyl glycerol	ج	
cardiolipin	د	
منبع:		



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Biochemistry

Which of the following is a nonaromatic amino acid with hydroxyl R-group?

50

Phenylalanine	الف
Lysine	ب
Threonine	ج
Methionine	د

منبع:

Biochemistry

Which among the following is the structure of myoglobin?

51

Homodimer	الف
Monomer	ب
Heterodimer	ج
Tetramer	د

منبع:

Biochemistry

A vitamin derived from amino acid is:

52

Biotin	الف
Niacin	ب
Folic acid	ج
Pantothenic acid	د

منبع:

Biochemistry

Which of the following is produced with the combination of apoenzyme and coenzyme:

53

Enzyme substrate complex	الف
Prosthetic group	ب
Enzyme substrate complex	ج
Holoenzyme	د

منبع:

Biochemistry

All statements are true regarding carbohydrates with the exception of:

54

Galactose and glucose are epimers of each other	الف
Amino sugars can be found in the structure of erythromycin.	ب
N-glycosidic bonds can be found in the structure of ATP.	ج
Amylose is a branched polysaccharide	د

منبع:



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Biochemistry

Which statement is true about heparin:

55

Heparin is a phosphorylated aglycon الف

Heparin is a proteoglycan ب

It can be used for the storage and process of blood products ج

It is a storage polysaccharide د

منبع:

Biochemistry

Which statement is true about buffers:

56

Buffers are mixtures of strong acids and their conjugate base. الف

Buffers tend to resist changes in pH when large amounts of acid or base are added ب

The Henderson-Hasselbalch equation is important for understanding buffering system ج

Hemoglobin can act a urine buffer د

منبع:

Biochemistry

In which pH, a hypothetical aqueous solution can act as an effective buffer?

57

$pH = pK_a \pm 1$. الف

$pH = pK_a \pm 2$. ب

$pH = K_a \pm 2$. ج

$pH = K_a \pm 1$. د

منبع:

Physiology: Dr. Ranjbaran

Cell Physiology		Question time (seconds):
58	The intracellular fluid contains	→
1	large amounts of phosphate ions	
2	large amounts of chloride ions	
3	large amounts of bicarbonate ions	
4	large amounts of fatty acids	

20.24 x 30